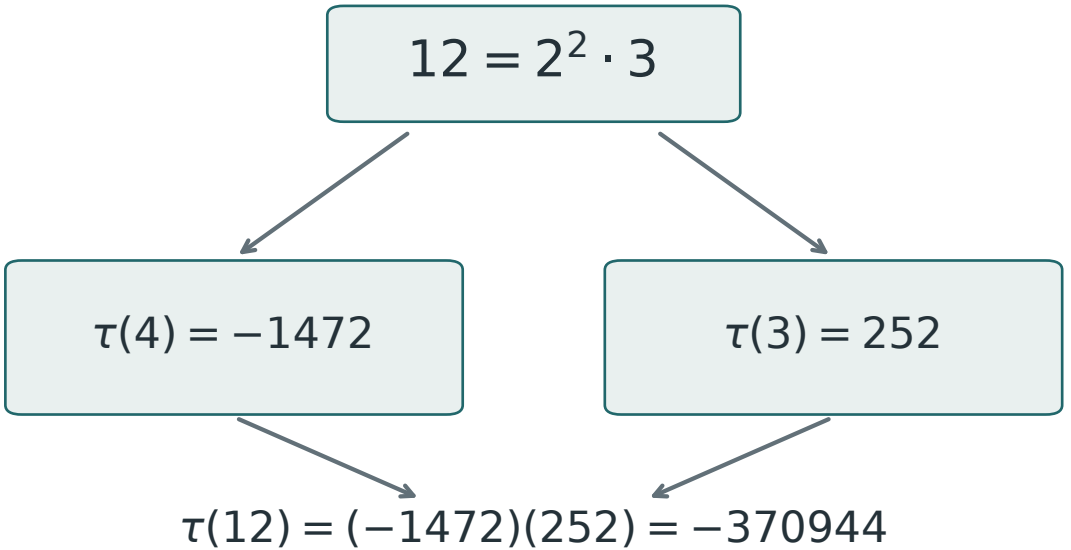
$12 = 2^2 \cdot 3$
$\tau(4) = -1472$
$\tau(3) = 252$
$\tau(12) = (-1472)(252) = -370944$